

2.2. Which of the following flows are physically possible, that is, satisfy the continuity equation? Substitute the expressions for density and for the velocity field into the continuity equation to substantiate your answer.

(a) Water, which has a density of 1.0 g/cm^3 , is flowing radially outward from a source in a plane such that $\vec{V} = (K/2\pi r)\hat{e}_r$. Note that $v_\theta = v_z = 0$. Note also that, in cylindrical coordinates,

$$\nabla = \hat{e}_r \frac{\partial}{\partial r} + \frac{\hat{e}_\theta}{r} \frac{\partial}{\partial \theta} + \hat{e}_z \frac{\partial}{\partial z}$$

(b) A gas is flowing at relatively low speeds (so that its density may be assumed constant) where

$$u = -\frac{2xyz}{(x^2 + y^2)^2} U_\infty L$$

$$v = \frac{(x^2 - y^2)z}{(x^2 + y^2)^2} U_\infty L$$

$$w = \frac{y}{x^2 + y^2} U_\infty L$$

Here U_∞ and L are a reference velocity and a reference length, respectively.

2.3. Two of the three velocity components for an incompressible flow are:

$$u = x^2 + 2xz \quad v = y^2 + 2yz$$

What is the general form of the velocity component $w(x,y,z)$ that satisfies the continuity equation?

2.4. A two-dimensional velocity field is given by

$$u = -\frac{Ky}{x^2 + y^2} \quad v = +\frac{Kx}{x^2 + y^2}$$

where K is a constant. Does this velocity field satisfy the continuity equation for incompressible flow? Transform these velocity components into the polar components v_r and v_θ in terms of r and θ . What type of flow might this velocity field represent?